

Nonzero irreducibility without a unit assumption

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible`

def

Requires nonzero action as well as the absence of proper closed reducing subspaces.

Statement

Let A be a complex C^* -algebra, with no unit assumed, H a complex Hilbert space, and $\pi : A \rightarrow \mathcal{B}(H)$ a $*$ -representation. The following predicate defines nonzero irreducibility.

Definition

The first condition is

$$\exists a \in A : \pi(a) \neq 0.$$

The second says: for every complex linear subspace $L \subseteq H$ which is closed and satisfies

$$\pi(a)x \in L, \quad \pi(a)^*x \in L \quad (a \in A, x \in L),$$

one has $L = 0$ or $L = H$. These are the definitions of nonzero action and reduction in Nonzero action and closed reducing subspaces. Because $\pi(a)^* = \pi(a^*)$ and A is closed under adjoints, invariance under every $\pi(a)$ also gives the stated adjoint invariance; the source records both explicitly.

Assumptions

No separability or injectivity is assumed. The nonzero requirement is a property of the action π , not merely of its underlying Hilbert space.

Conclusion

The zero representation is excluded, and every closed reducing complex subspace is either 0 or H .

Main citations

- Exact definition and context
- Nonzero action and closed reducing subspaces
- Representations of possibly non-unital algebras

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
def IsIrreducible (pi : NonUnitalCStarRepresentation A H) : Prop :=
  pi.IsNonzero  $\wedge$   $\forall$  L : Submodule  $\mathbb{C}$  H, pi.Reduces L  $\rightarrow$  L =  $\perp$   $\vee$  L = T
```

In the source	Mathematical meaning
(pi : NonUnitalCStarRepresentation A H)	The given action of A by bounded operators on H .
: Prop	The output is a condition on that action.
pi.IsNonzero	Some $a \in A$ has $\pi(a) \neq 0$ as an operator. The condition $H \neq 0$ alone would not suffice.
$\forall$ L : Submodule $\mathbb{C}$ H	For every complex linear subspace L of H .
pi.Reduces L	L is norm closed and, for all $a \in A$ and $x \in L$, both $\pi(a)x$ and $\pi(a)^*x$ belong to L .
L = $\perp$ $\vee$ L = T	That same reducing subspace must be the zero subspace or all of H .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible
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